

+CO Creative Coding

AI's Hidden Water Footprint

For the mid-term presentation on Monday, May 18th, we decided to showcase the initial visuals for our project, "AI's Hidden Water Footprint". Our main focus so far has been on the coding; we will develop the audio aspect in the next phase of the module.

I wanted to generate water effects by programming, starting with the typography. My goal was to create a visual illusion where the text appears to dissolve into water. Initially, I scattered the typography across the screen using a function that generates text at the cursor's location upon a mouse click. When overlapping in the same position, it created a shimmering effect reminiscent of moving water.

To build on this, I consulted Gemini to implement a blending effect for the text. Utilizing a sine wave made it possible to smoothly transition the effects and cause the typography to fade away.

Alternatively, I experimented with a code snippet that simulates rain. While rain is not directly linked to the water consumption of data centers, it allowed me to explore different interactive elements. The digital rain reacts to the cursor, changing speed and shifting direction from left to right based on user input.

These two snippets are our first prototypes for the final performance. While there is still a lot of trial and error involved, successfully coding a water-like effect feels like a great first step.

Alongside the technical work, I researched the topic further. Even though I was aware of the environmental impact of data centers, the statistics were startling.

Official data on water consumption remains scarce. However, the International Energy Agency estimates current demand at around

560 billion liters per year, which could escalate to 1,200 billion liters by 2030—a volume equivalent to the annual water needs of roughly ten million households.

Our next challenge is to integrate these massive figures into our code in a way that captivates the audience. We will continue our research and look for interviews to make the underlying issue clearer and, above all, visually striking.